NBA Players Trajectories

EE-452: Network Machine Learning

EPFL - Spring 2025/2026

1 Introduction

Human motion forecasting aims to predict the future trajectories and poses of humans given observations of their past movements. Sports players' trajectories are a good example of such a setting. In this project, we will focus on basketball players from the National Basketball Association (NBA). The player and ball trajectories exhibit complex patterns of coordinated movement, rapid changes in direction, and strong interaction dynamics between multiple agents, providing a high-dimensional benchmark for studying multi-agent motion prediction and learning interaction-aware representations.

However, forecasting human motion in multi-agent environments presents several key challenges. First, human motion is inherently stochastic: multiple plausible futures may exist for the same observed history. Furthermore, interactions between agents introduce strong dependencies, where the motion of one individual influences and is influenced by the behavior of others. Finally, modeling data with high dimensionality and temporal structures requires models that can capture both spatial relationships and long-range temporal dependencies.

Graph-based deep learning models are a powerful framework for representing structured interactions. By modeling entities as nodes and relationships as edges, graph neural networks (GNNs) enable the propagation of information across the graph, allowing the model to capture dependencies between agents while jointly modeling temporal evolution. Graph-based methods thus provide a flexible way to learn interaction-aware representations directly from data, making them particularly well-suited for multi-agent motion prediction scenarios.

Main Objective: This project aims to explore the utility of graph-based methods for human motion forecasting, and more precisely NBA players' trajectories. Students should compare graph-based and non-graph-based approaches, evaluating their advantages and limitations. Through this comparison, students will gain insights into how graph structures influence model performance and interpretability. These insights should be clearly articulated in the project report and presentation.

2 Dataset & Task Definition

In this project, we use an alternate version of the NBA highlight dataset from the SocialVAE paper [4]. It consists of more than 6K highlight plays from the 2015-2016 season. The player and ball movements have been recorded with a fixed frame rate for a variable number of frames, ranging from 20 to more than 100 steps. An example of such a trajectory is shown in Figure 1.

Dataset: The data is stored as PyTorch tensors ($.pt$) with dimension $L \times N \times D$, where L is the sequence length, N the number of entities and D the feature dimension. While N and D are fixed for the whole dataset, the length of sequences L may vary among samples. The features are the concatenation of the spatial coordinates, a binary variable *isplayer* indicating whether the entity is a player or the ball, and finally a variable *team* encoding the team membership (-1 = Team A, 0 = Ball, 1 = Team B). The final feature vector is thus: $[x,\ y,\ isplayer,\ team]$. Although *isplayer* and *team* are redundant, both are provided to facilitate filtering. Since these fields are not used for evaluation, they can be dropped during pre-processing if convenient. Note that only the spatial coordinates are dynamic features, whereas the others are static over time.

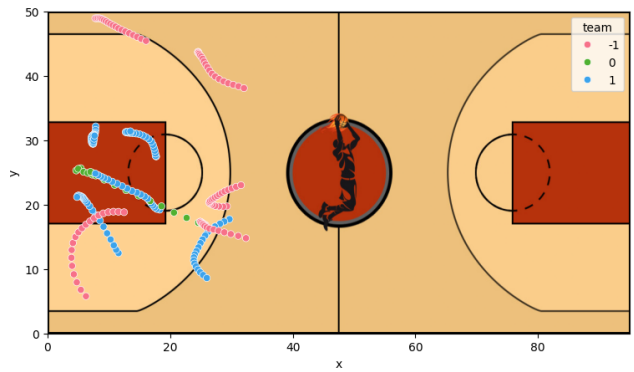

Figure 1: Illustration of the trajectory of players and the ball during one highlight. The players are associated with team -1 or 1, the ball with group 0.

Objective: The goal of the project is, given a context sequence of length C, to predict the trajectory of each entity (players and ball) for a horizon of length H. In our setup, we define $C = 8$ and $H = 12$, which gives an overall window of 20 time steps. Note that the window size is smaller than most sequences. You can play with it to sample different windows inside each sequence. The evaluation metric that we will use is the ADE (average discrepancy error), defined as the average Mean Square Error (MSE) over the predicted time steps in the horizon H:

$$\text{ADE} = \frac{1}{H \times N} \sum_{t=C+1}^{C+H} \sum_{n} \text{MSE}(\hat{z}_t^n, z_t^n), \tag{1}$$

where $\hat{z}_t^n = (\hat{x}_t^n, \hat{y}_t^n) \in \mathbb{R}^2$ corresponds to the predicted position of the entity n (player or ball) at timestep t, and $z_t^n = (x_t^n, y_t^n) \in \mathbb{R}^2$ to the corresponding ground truth. Therefore, the evaluation metric is computed only on the spatial coordinates (x, y), not on static features.

Intended Approach: For this task, graph-based representations can be particularly useful to capture spatial properties, interactions between players, and the heterogeneity of entities on the field. The students are encouraged to explore different graph constructions, drawing inspiration from both the geometry and semantics of the problem. Importantly, we expect all proposed solutions to include a graph-based component as a core element of the methodology. Students may optionally compare their approach against alternative baselines, such as vision-based methods, but submissions that do not incorporate a graph-based solution will not be considered aligned with the objectives of the project.

3 Kaggle competition

To enhance engagement and encourage experimentation, we will host a Kaggle competition where students can test their baselines in a competitive setting. This competition provides an opportunity to benchmark models and refine approaches. Additionally, to ensure a meaningful challenge, we will award a bonus to the top-performing groups. See Section 7 for more details.

While the Kaggle leaderboard provides a way to compare performance across teams, it should primarily serve as a reference rather than the sole evaluation metric. We strongly encourage groups to conduct and report local cross-validation performances using a validation set within the provided training data. This ensures that model selection is not solely drive by Kaggle rankings but is guided by robust internal evaluation practices.

4 Provided Pipeline

We provide a python notebook associated with this PDF to help you start with the project. It contains visualization tools and a basic training pipeline to train a naive temporal baseline, without graph prior, whose score is highlighted as *solution.csv* on the Kaggle leaderboard. You are invited to

take inspiration from this pipeline or to design your own! Many improvements can be made, e.g., on the sampling process, model architecture or training procedure, among others. The NBA dataset has been used in many different studies ([4, 3, 1, 2]), do not hesitate to read them to gather some ideas on how to design an interesting architecture and transfer their modeling to graph methods!

5 Computational Resources

All enrolled students will have access to GPU resources on the SCITAS cluster, Izar. If you are not familiar with SCITAS, please review the documentation available at SCITAS Documentation. The dataset (approximately 1 GB) will be available on the cluster. Alternatively, you may choose to use your own computational resources or Google Colab.

6 Project Expectations & Deliverables

Students are expected to submit **three deliverables: a report, a presentation, and the corresponding code**.

6.1 Code

We recommend using libraries already utilized in the exercise session notebooks (such as *PyTorch*, *PyTorch Geometric*, *NetworkX*, ...), but students are free to use any existing open-source code to build their models. The code should be well-documented and reproducible.

In addition to the code submission, students must provide a short screen-recorded video with voice narration presenting their implementation (**maximum duration: 5 minutes**). The recording should briefly explain the structure and organization of the codebase, highlight clarity and documentation, and include a short demonstration showing the code running. Students may use any screen recording software of their choice. The video must be submitted in MP4 format.

6.2 Report

The report should follow the official ICLR 2026 format. Each group must submit a strictly 4-page report, excluding references, following a structured format similar to a scientific paper. The report should include the following sections:

- **Introduction**: Problem motivation and objectives.
- **Method**: Approach and technical details.
- **Results**: Experiments, findings, and discussion.
- **Conclusion**: Summary and insights.

6.3 Presentation

Each group will give a presentation of up to 10 minutes, summarizing their work. The structure of the presentation should closely follow the report, covering the problem statement, methodology, key findings, and conclusions. A 5-minute Q&A will follow the presentation.

6.4 Submission Deadlines

- **Report and Code Submission**: Due on Moodle by May 29th, 2026.
- **Presentation Dates**: June 15th–16th, 2026.

7 Evaluation Criteria

The project grade will be based on three components: presentation, code, and report. The grading breakdown is shown in Table 1. All team members will receive the same grade.

7.1 Bonus Points

We will award bonus points to the top 10 performing teams in the Kaggle competition. The bonus will be distributed in a linearly spaced manner, with the 1st team receiving the maximum bonus of 0.5 and the 10th team receiving the minimum of 0.05, with intermediate teams awarded proportionally (linearly) decreasing bonuses. These bonus points will be added to the final project grade (capped at 6).

Category	Criterion (%)	Poor (1)	Limited (2)	Satisfactory (3)	Good (4)	Excellent (5)
Presentation and Report (80%)	Description of methods (20%)	Incorrect or missing	Incomplete explanation	Basic description	Mostly clear with minor gaps	Clear, precise, technically sound
	Originality of the approach (10%)	No baselines	Limited novelty (just baselines)	Standard approach	Some originality	Highly original
	Interpretation of results (25%)	Misinterpretation	Superficial comments	Basic discussion	Good interpretation	Deep and critical analysis
	Presentation: communication and clarity (15%)	Unclear	Hard to follow	Understandable	Clear overall	Very clear and engaging
	Report: writing quality and overall presentation (10%)	Poorly written	Many issues	Acceptable writing	Well written	Excellent structure and writing
Code (20%)	Functionality (10%)	Does not run	Major problems	Runs with fixes	Minor issues	Runs without issues
	Code quality and documentation (10%)	Poor quality	Hard to read	Adequate	Mostly clean	Very clean and well documented

Table 1: Grading breakdown.

References

[1] Fu, Y., Yan, Q., Wang, L., Li, K., Liao, R.: Moflow: One-step flow matching for human trajectory forecasting via implicit maximum likelihood estimation based distillation. In: Proceedings of the Computer Vision and Pattern Recognition Conference. pp. 17282–17293 (2025), https://arxiv.org/abs/2503.09950

[2] Gao, Y., Luan, P.C., Messaoud, K., Feng, L., Alahi, A.: Omnitraj: Pre-training on heterogeneous data for adaptive and zero-shot human trajectory prediction. arXiv preprint arXiv:2507.23657 (2025), https://arxiv.org/abs/2507.23657

[3] Sestak, F., Toshev, A., Fürst, A., Klambauer, G., Mayr, A., Brandstetter, J.: Lam-slide: Latent space modeling of spatial dynamical systems via linked entities. arXiv preprint arXiv:2502.12128 (2025), https://arxiv.org/abs/2502.12128

[4] Xu, P., Hayet, J.B., Karamouzas, I.: Socialvae: Human trajectory prediction using timewise latents. In: European Conference on Computer Vision. pp. 511–528. Springer (2022), https://arxiv.org/abs/2203.08207